

MSc Individual Research Project

Causal Structure Discovery via a Self-Implemented FCI⁺ Algorithm

A Case Study on the Tennessee STAR Project

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Abstract

Constraint-based causal discovery is attractive when a researcher does not want to commit to a single pre-specified directed acyclic graph, but classical methods such as PC assume that all relevant common causes are observed. This report develops, validates, and applies a self-implemented Python package for Fast Causal Inference (FCI) and its sparse extension FCI⁺. Both methods allow latent confounding and selection effects and return a Partial Ancestral Graph (PAG), which encodes the causal features shared by a Markov equivalence class rather than asserting one unique causal DAG.

The software contribution is an auditable implementation containing PAG data structures, continuous, discrete, nonlinear, and missing-value conditional independence tests, standard FCI Possible-D-SEP refinement, the hierarchical D-SEP search of FCI⁺ Algorithm 2, orientation rules, background knowledge, bootstrap stability analysis, typed public APIs, and machine-readable diagnostics. Exact m-separation tests recover three published reference PAGs. On the central Figure 4(b) example of Claassen et al. (2013), FCI⁺ recovers the exact PAG with 63 conditional-independence queries, compared with 102 for standard FCI. A seeded finite-sample experiment also shows the important qualification: FCI⁺ retains one extra edge at $N = 5,000$ but reaches exact recovery at $N = 50,000$. Algorithmic correctness under an oracle therefore does not eliminate conditional-independence testing error in real data.

The application is deliberately separated from the reusable algorithm package. It uses the official Tennessee Student/Teacher Achievement Ratio (STAR) data: 11,601 student records and 379 variables, with a kindergarten analysis cohort of 6,325 students across 79 schools. Three discrete analysis panels examine grade-3 observation, longitudinal achievement, and the focused class-assignment/grade-3 relationship. Descriptive school-cluster bootstrap comparisons show that small classes exceed regular classes by 13.9 score points at the end of kindergarten (95% interval 5.2 to 22.1) and by 11.7 points at grade 3 among observed students (3.7 to 19.4). The regular-class-with-aide contrasts are close to zero.

Every primary panel is first fitted by three paper-comparison algorithms: self-implemented standard FCI, self-implemented paper-profile FCI⁺, and CRAN `pcalg::fciPlus` as an independent R reference. The same compact-table G-square decision rule is supplied to all three so that the comparison is not driven by `pcalg`'s sparse-table default. The focused panel yields an exact three-way match: all algorithms return the same 10-edge PAG, including $K_Class \leftrightarrow Grade3_Achievement$. That edge nevertheless disappears for both self-implemented algorithms when α is reduced from 0.05 to 0.01. The attrition and longitudinal skeletons show substantial cross-implementation overlap, but several endpoint marks disagree and both FCI⁺ implementations produce at least one order- or chronology-sensitive endpoint. The self-implemented FCI⁺ uses 35–64% of the CI tests required by self-implemented standard FCI, but it is not uniformly the most plausible STAR graph. A fourth, explicitly separate robust application fit uses stable skeleton deletion, strongest-at-depth separating sets, conservative collider handling, and cautious orientation. It is exactly invariant across every cyclic variable order in all three panels and contains no fully directed temporal reversal, at the cost of more CI calls and less endpoint specificity.

Executive Summary

Research question. Can an independently implemented FCI⁺ algorithm recover auditable causal structure in the presence of latent variables, and what does its application add to the interpretation of the Tennessee STAR randomized class-size experiment?

Software outcome. The project produced an installable Python package with standard FCI, paper-aligned and practical FCI⁺ profiles, multiple CI tests, PAG diagnostics, stability analysis, HTML/JSON/CSV exports, strict type checking, and a Python 3.9–3.13 continuous-integration matrix. At the time of this report, all 351 repository tests pass.

Algorithm outcome. Exact oracle tests establish that the implemented search and orientation logic can recover the intended PAGs when conditional independence answers are correct. The published Figure 4(b) D-SEP link is removed with the separator $\{U, V, Z\}$, and FCI⁺ uses 39 fewer oracle queries than FCI on that fixture. Finite-sample simulations show that this efficiency does not guarantee equal empirical reliability at every sample size.

Independent implementation outcome. CRAN `pcalg` 2.7-12 under R 4.5.3 was run on the same encoded panels with a Python-equivalent G-square test. It is treated as a differential reference, not ground truth. The focused PAG is exactly identical across all three algorithms. In the attrition and longitudinal panels, pairwise skeleton Jaccard values range from 0.765 to 0.929, but exact endpoint agreement on shared edges is lower. Cyclic variable orders preserve every `pcalg` skeleton yet reproduce the original PAG endpoints in only 67–78% of orders.

Robust application outcome. The paper implementations remain unchanged for fidelity testing. A separate practical profile is used for empirical interpretation. Across all 26 cyclic column shifts it reproduces each panel’s complete PAG exactly, retains the main target adjacencies in every order, and replaces chronology-conflicting definite arrows with unresolved circle endpoints. This improves stability without claiming new causal identification.

STAR outcome. The official data were converted into three independent application panels:

1. an attrition panel containing 5,744 students and a grade-3 observation indicator;
2. a longitudinal panel containing 2,787 complete cases and both kindergarten and grade-3 achievement;
3. a focused complete-case panel containing 2,976 students, excluding kindergarten achievement from the discovery node set.

The strongest result does not come from a PAG endpoint. It comes from the randomized-arm comparison: small classes have higher observed kindergarten scores than regular classes, whereas adding an aide to a regular class does not show a comparable advantage. The later grade-3 contrast is also positive among observed students, but attrition prevents the same strength of causal interpretation.

Causal interpretation.

- The randomized design, together with the positive school-cluster interval, supports a beneficial causal effect of small classes on early achievement for students with observed kindergarten outcomes.
- There is no corresponding evidence that a regular class with an aide improves achievement relative to a regular class.

- Kindergarten achievement is the most stable structural correlate of grade-3 achievement and grade-3 observation.
- The focused bidirected treatment-outcome edge is replicated exactly by all three algorithms, but is threshold-sensitive and cannot override knowledge of randomization.
- Cross-implementation skeleton consensus is more credible than endpoint marks that conflict with chronology or change under variable order.
- The robust application profile supports the same main adjacencies but leaves their directions unresolved; this is the preferred finite-sample reading of the discovery output.
- PAG endpoints should be read as equivalence-class constraints, not regression coefficients or direct effects.

Bottom line. The case study demonstrates practical use of the algorithm and reveals where selection and modelling choices matter. It does not convert exploratory causal discovery into a substitute for design-based effect estimation.

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Abbreviations and Notation

CI	Conditional independence.
DAG	Directed acyclic graph.
MAG	Maximal ancestral graph, representing observed-variable relations after latent variables or selection variables have been marginalised or conditioned upon.
PAG	Partial Ancestral Graph, representing the invariant features of a Markov equivalence class of MAGs.
FCI	Fast Causal Inference.
FCI⁺	Sparse FCI extension based on a logical characterisation of D-SEP links and hierarchical separating sets.
STAR	Tennessee Student/Teacher Achievement Ratio experiment.
$X \perp Y \mid S$	X and Y are conditionally independent given conditioning set S .
$X \circ\!\!\!\circ Y$	Neither PAG endpoint is identified.
$X \circ\!\!\!\rightarrow Y$	Y is not an ancestor of X , while the endpoint at X remains unresolved.
$X \leftrightarrow Y$	Arrowheads at both endpoints; compatible with latent confounding but not proof of one specific hidden cause.
$X \rightarrow Y$	An invariant ancestral relation in the PAG; not necessarily a direct causal effect.

Part I

Algorithm, Implementation, and Validation

Introduction

1.1 Motivation

Many empirical studies ask causal questions in settings where not every common cause has been measured. Family background, institutional selection, health, motivation, and administrative recording processes may all generate dependence that cannot be represented honestly by a causally sufficient DAG over the recorded columns. Fitting a directed graph while silently assuming causal sufficiency can turn uncertainty into unjustified arrows.

FCI was designed for this problem. Under causal Markov, faithfulness, and appropriate sampling assumptions, it uses conditional-independence information to learn a PAG that remains valid when latent confounders and selection variables may exist (Spirtes et al., 2000; Richardson and Spirtes, 2002). The price of this broader model class is additional search and greater endpoint ambiguity. In particular, standard FCI’s Possible-D-SEP refinement may test many separating sets that are not contained in local adjacency neighbourhoods.

FCI⁺ addresses the search problem for sparse causal graphs. Claassen et al. (2013) provide a logical characterisation of D-SEP links and a hierarchical procedure for finding the relevant separating sets. For a fixed maximum degree k , their result yields a polynomial worst-case bound in the number N of observed variables, commonly written

$$O\left(N^{2(k+2)}\right).$$

This is a sparsity-dependent guarantee, not a statement that every finite dataset will be easy or statistically well behaved.

1.2 Project contributions

The project makes two distinct contributions.

1. **Algorithm contribution.** A self-contained Python implementation of FCI and FCI⁺, with multiple CI tests, explicit PAG endpoint operations, paper-aligned profiles, practical finite-sample profiles, diagnostics, stability selection, type annotations, and release validation.
2. **Application contribution.** A separate Tennessee STAR case study that applies the public algorithm API without importing STAR-specific assumptions into the package. It uses randomized-arm contrasts as an external design-based reference and PAG learning as a structural sensitivity analysis.

This separation is central to the research design. An algorithm should remain reusable and domain-neutral; an application must state its own coding, selection, measurement, clustering, and interpretation decisions.

1.3 Research questions

The report addresses four questions:

- RQ1.** Can the implementation recover published PAG structures under exact conditional-independence information?
- RQ2.** Does the FCI⁺ hierarchical D-SEP stage reduce search work relative to standard FCI on controlled examples and the STAR application?
- RQ3.** Which STAR relationships are stable across FCI, FCI⁺, and school-cluster resampling?
- RQ4.** Which causal conclusions are supported by experimental design, and which remain only exploratory PAG interpretations?

Causal-Graphical Background

2.1 Conditional independence and graphical assumptions

Let V be the set of observed variables, L latent variables, and S selection variables. Constraint-based discovery begins from statements of the form

$$X \perp Y \mid Z,$$

where $Z \subseteq V \setminus \{X, Y\}$. A causal graph and probability distribution are linked by two central assumptions.

Causal Markov condition. Graphical separation implies probabilistic conditional independence.

Faithfulness. Probabilistic conditional independences arise from graphical separation rather than exact cancellation of causal effects.

Under causal sufficiency these assumptions support DAG-based methods such as PC. When latent or selection variables are present, the appropriate observed-variable object is an ancestral graph or MAG (Richardson and Spirtes, 2002). Different MAGs may encode the same conditional-independence model, so the identifiable output is a PAG rather than one unique graph.

2.2 Meaning of PAG endpoints

Table 2.1: Interpretation of common PAG edge marks.

Mark	Interpretation
$X \text{ o--o } Y$	The adjacency is retained, but neither ancestral endpoint is identified.
$X \text{ o--> } Y$	The arrowhead at Y excludes Y as an ancestor of X ; the endpoint at X remains unresolved.
$X \text{ --> } Y$	The tail-arrowhead pattern supports X as an ancestor of Y through some path in every represented MAG. It does not assert that the edge is a direct structural coefficient.
$X \text{ <--> } Y$	Both endpoints are arrowheads. The pattern is compatible with latent confounding, but it does not identify a unique latent variable and can be distorted by selection or CI-test error.
$X \text{ - } Y$	An undirected ancestral-graph edge, often associated with selection structure.

The distinction between ancestry and direct effect is especially important. FCI-family algorithms identify equivalence-class features. They do not estimate causal effect magnitudes, adjustment sets, or intervention distributions without additional analysis.

2.3 Standard FCI

The implemented paper-aligned FCI pipeline follows the broad sequence in Spirtes et al. (2000):

1. start from a complete undirected graph over observed variables;
2. remove adjacencies using CI tests over progressively larger local conditioning sets, recording separating sets;
3. orient unshielded colliders using those separating sets;
4. form the partially oriented graph used to compute Possible-D-SEP candidate sets;
5. search those larger candidate sets and remove further adjacencies;
6. re-orient colliders and propagate sound PAG orientation rules to closure.

Possible-D-SEP is required because a valid separator for two variables need not remain adjacent to either endpoint after the initial PC-like skeleton stage. This is precisely the situation that makes FCI more robust to latent variables and potentially more expensive than PC.

2.4 FCI⁺ and hierarchical D-SEP

FCI⁺ replaces the standard Possible-D-SEP refinement with a sparse search based on D-SEP links. A D-SEP link is an adjacency that survives local skeleton search even though the variables are separable

by a set involving more distant nodes. The paper shows that such links can be recognised through local logical patterns in an augmented skeleton and that their separating sets can be built hierarchically.

The implementation’s paper profile follows Algorithm 2:

1. run adjacency search to depth k ;
2. use minimal separating sets to identify collider information;
3. construct the augmented skeleton;
4. enumerate candidate D-SEP links implied by the augmented structure;
5. build hierarchical bases from the two sides of each candidate edge;
6. test combinations up to the fixed sparsity bound k ;
7. remove confirmed D-SEP links and repeat as required;
8. apply the full implemented PAG orientation schedule.

The same k is used for ordinary conditioning depth and the hierarchical D-SEP base in the literal paper profile. The practical profile allows the conditioning cap and sparsity bound to be expressed separately, because they play different operational roles.

Interpretation caution. The polynomial FCI⁺ result assumes sparsity, faithfulness, and correct CI information. In an empirical dataset the true maximum MAG degree is unknown, and a finite-sample CI error may prevent the augmented skeleton from exposing the candidate D-SEP link.

Software Design and Implementation

3.1 Package architecture

The reusable implementation is located under `src/fci_engine`. The STAR application is outside this package. Table 3.1 summarises the main modules.

Table 3.1: Implementation architecture.

Module	Responsibility
<code>graph/</code>	PAG storage, endpoint mutation, adjacency operations, edge representations, and graph validation.
<code>ci/</code>	Public CI-test interface, Fisher-Z, missing-value Fisher-Z, chi-square, G-square, kernel CI, and CI-result caching.
<code>discovery/skeleton.py</code>	PC-style adjacency search, conditioning-depth control, stable deletion option, and separating-set selection.
<code>discovery/pdsep.py</code>	Standard FCI Possible-D-SEP candidate generation and skeleton refinement.
<code>discovery/dsep.py</code>	FCI ⁺ augmented skeleton, D-SEP candidate recognition, hierarchy construction, and duplicate-test avoidance.
<code>discovery/rules.py</code>	Collider orientation and PAG orientation rules, including discriminating-path logic.
<code>knowledge/</code>	Required and forbidden background-knowledge constraints.
<code>metrics/</code>	Oracle accuracy, reference benchmarks, and bootstrap stability selection.
<code>simulation/</code>	Published MAG fixtures, exact m-separation oracle tests, and seeded SEM generators.
<code>result.py</code>	Result summaries, edge explanations, assumptions, diagnostics, and JSON/CSV/HTML artifact export.

Shared orchestration in `discovery/base.py` keeps data preparation, CI-test selection, metadata, tracing, and final result construction consistent between standard FCI and FCI⁺. This reduces duplicated logic while preserving separate algorithm entry points.

3.2 Conditional-independence tests

The package exposes a common CI-test protocol so that graph search is not tied to one data type.

- **Fisher-Z** for continuous, approximately multivariate Gaussian data without missing values.
- **Missing-value Fisher-Z** using test-specific complete rows.
- **Chi-square and G-square** for discrete contingency tables.
- **Kernel CI** for nonlinear continuous relationships.
- **Custom tests** through the same public interface.

Repeated calls are cached by the unordered variable pair and canonicalised conditioning set. This matters because both Possible-D-SEP and hierarchical D-SEP searches can revisit logically equivalent tests from different paths.

3.3 Finite-sample profiles

Table 3.2: Public analysis profiles and their intended use.

Profile	Main settings	Intended use
<code>fci(..., profile="paper")</code>	Immediate skeleton updates, first separating set, unbounded depth unless overridden, standard Possible-D-SEP and orientation schedule.	Reproduction of the traditional FCI search schedule.
<code>fci_plus(..., profile="paper", k=...)</code>	Algorithm 2 settings, shared k bound, first separating set, no standard Possible-D-SEP stage, standard orientation.	Paper-aligned validation and algorithm comparison.
<code>fci_plus(..., profile="practical")</code>	Stable skeleton, strongest-at-depth separating set, bounded search, conservative/robust orientation, optional automatic alpha.	Exploratory finite-sample work, subject to sensitivity analysis.

The STAR paper comparison uses the paper profiles for both local algorithms so that it concerns the published search strategies rather than different finite-sample defaults. A separate STAR application result uses `profile="practical"`. This preserves algorithm fidelity while applying additional safeguards to conclusions drawn from finite-sample data.

3.4 Public API and artifacts

Listing 3.1: Paper-aligned FCI+ public API.

```
from fci_engine import fci_plus
from fci_engine.ci import GSquareTest

result = fci_plus(
    data,
    profile="paper",
    k=3,
    alpha=0.05,
    ci_test=GSquareTest(alpha=0.05),
)

print(result.summary())
print(result.to_pandas_edges())
result.save_artifacts("output", stem="analysis")
```

Listing 3.2: Recommended finite-sample FCI+ application API.

```
robust_result = fci_plus(
    data,
    profile="practical",
    max_cond_set_size=3,
    sparsity_bound=3,
    alpha=0.05,
    ci_test=GSquareTest(alpha=0.05),
)
```

An `FCIResult` stores the learned PAG together with the resolved configuration, sample and node counts, separating sets and their sources, CI trace, orientation trace, D-SEP diagnostics, runtime, and assumption notes. This is important for auditability: a plotted edge should not be detached from the settings and decisions that produced it.

3.5 Engineering quality

The package targets Python 3.9 through 3.13 and includes a `py.typed` marker. Strict MyPy checks the package and the STAR application. Ruff checks style and formatting. GitHub Actions independently runs Pytest across all five supported Python versions, then runs the quality checks and builds, installs, and imports the wheel in a clean environment. The revision used for the STAR case study passed all jobs and all 351 local tests.

These checks establish software consistency; they do not establish the truth of empirical causal conclusions. Statistical validation requires separate oracle and finite-sample evidence.

Algorithm Validation

4.1 Exact oracle tests

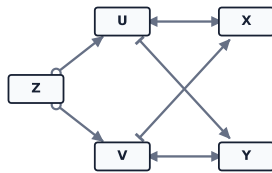
The strongest algorithm tests replace sample-based CI decisions with exact m-separation in a known MAG. This isolates graph-search correctness from sampling error. The regression suite contains three published structures:

1. the canonical Figure 4(b) D-SEP example from Claassen et al. (2013);
2. a latent-variable reference graph derived from the FCI literature (Spirtes et al., 2000);
3. an orientation-rule reference graph associated with completeness results in Zhang (2008).

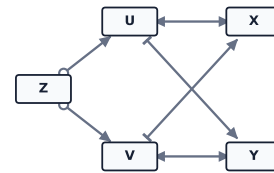
Both FCI and FCI⁺ recover the complete target PAG for all three oracle fixtures. Figure 4(b) is additionally tested under four variable orders.

Exact m-separation isolates algorithmic correctness from sampling error

Published Figure 4(b) oracle PAG



**Recovered FCI+ PAG
63 CI queries; exact endpoint match**



The X-Y candidate is removed by the hierarchical D-SEP stage using the separating set {U, V, Z}.

Figure 4.1: Exact-oracle validation on the central Figure 4(b) example. The learned FCI⁺ PAG matches every oracle adjacency and endpoint.

The initial adjacency stage retains the $X - Y$ link. FCI⁺ identifies it as a D-SEP candidate and removes it using $\{U, V, Z\}$. Standard FCI reaches the same final PAG through Possible-D-SEP. The measured query counts on this fixture are:

Table 4.1: Paper-aligned Figure 4(b) oracle comparison.

Algorithm	CI queries	Difference	Final PAG
Standard FCI	102	reference	Exact
FCI+	63	-39 (-38.2%)	Exact

4.2 Finite-sample integration test

Exact CI answers are necessary for testing the algorithm, but real data replace them with noisy statistical decisions. A seeded latent linear SEM based on the same reference graph provides the following integration result:

Table 4.2: Seeded finite-sample recovery on the Figure 4(b) SEM.

Sample size	FCI+	Standard FCI	Interpretation
$N = 5,000$	Extra $X \leftrightarrow Y$; exact F1 = 0.923	Exact PAG; exact F1 = 1.000	The sample CI decisions do not expose the FCI+ D-SEP candidate.
$N = 50,000$	Exact PAG; exact F1 = 1.000	Exact PAG; exact F1 = 1.000	Both algorithms recover this seeded fixture; this is not a universal sample threshold.

4.2.1 Multi-seed robustness regression

The application safeguard was also tested on 15 deterministic latent-SEM cases: five realistic graph families, three random seeds each, and 2,500 observations per case. The oracle PAG is known for every case. This test compares the unchanged paper profile with the practical profile used in the STAR application.

Table 4.3: Mean recovery across 15 finite-sample oracle cases.

Profile	Skeleton F1	Exact F1	Endpoint acc.	Semantic F1	Compatible endpoint	CI calls
Paper	0.980	0.536	0.744	0.951	0.948	197.1
Robust	0.980	0.703	0.821	0.980	0.962	380.7

The robust profile preserves mean skeleton recovery while improving exact and compatibility-aware endpoint scores. It uses about 1.93 times as many CI calls. This does not prove universal dominance, but it provides a known-truth regression check that the STAR stability gain is not achieved by simply discarding useful structure.

This result is a useful warning against claiming that fewer CI tests imply uniformly better finite-sample accuracy. FCI and FCI⁺ can make different errors because they use different logical search routes after the initial skeleton.

4.3 Reference software

The STAR analysis runs CRAN `pcalg` 2.7-12 under R 4.5.3 as an independent reference implementation (Kalisch et al., 2012). The runner calls `pcalg::fciPlus` with selection-bias orientations enabled. It supplies a custom G-square test that exactly mirrors the Python compact-table rule: zero-marginal rows and columns are removed separately within each observed conditioning stratum, and the resulting likelihood-ratio statistics and degrees of freedom are summed. This avoids `disCItest`'s default minimum-cell shortcut, which would otherwise change the CI decisions rather than merely change the implementation.

The R output is treated as a differential reference, not ground truth. In particular, `pcalg::fciPlus` does not expose the local implementation's explicit $k = 3$ sparsity-bound parameter. Comparisons therefore report skeleton overlap, exact endpoint agreement, target-edge marks, runtime, CI calls, and cyclic-order sensitivity instead of awarding correctness by package name.

Key result. The oracle tests support the correctness of the implemented FCI and FCI⁺ search-and-orientation pipeline under exact conditional-independence information. The finite-sample tests simultaneously show that empirical results remain limited by CI-test error and tuning choices.

Part II

Tennessee STAR Application

Study Design and Data

5.1 The Tennessee STAR experiment

The Tennessee Student/Teacher Achievement Ratio project was a large randomized experiment conducted in kindergarten through grade 3. Students and teachers were assigned to small classes, regular classes, or regular classes with a full-time aide. The official technical report describes small classes of approximately 13–17 students and regular classes of approximately 22–25 students (Word et al., 1990). Subsequent economic analysis has used STAR as a central source of evidence on class size and educational production (Krueger, 1999).

The present analysis uses the official Harvard Dataverse release (Achilles et al., 2008). The student-level file contains 11,601 records and 379 variables and is released under CC0 1.0. Data codes were checked against the accompanying 146-page user guide.

5.2 Why use causal discovery on a randomized experiment?

Randomization already provides a stronger basis for estimating the class-size effect than an unconstrained structure-learning algorithm. The purpose of FCI is therefore not to rediscover or replace randomization. It is to examine:

- whether grade-3 observation is structurally related to early achievement or background variables;
- whether early and later achievement remain adjacent after conditioning on the measured context;
- how the learned PAG changes when an early post-assignment outcome is included or excluded;
- whether FCI and FCI⁺ agree on skeletons and important adjacencies;
- how much search work FCI⁺ saves on a real dataset.

This design also provides an unusually useful external audit. If a data-only PAG implies an implausible pre-treatment direction or suggests latent confounding of a genuinely randomized assignment, the discrepancy should be reported rather than accepted literally.

5.3 Cohort construction

The analysis begins with 6,325 students who have a kindergarten STAR class assignment, representing 79 kindergarten schools. Three panels are constructed for different questions.

Table 5.1: STAR analysis panels.

Panel	Rows	Nodes	Purpose
Attrition / observation	5,744	9	Relate kindergarten assignment, characteristics, context, and achievement to whether both grade-3 reading and mathematics scores are observed.
Longitudinal achievement	2,787	9	Study structure between kindergarten and grade-3 achievement among complete cases while retaining early achievement.
Focused treatment-outcome	2,976	8	Study the class-assignment/grade-3 relationship without allowing kindergarten achievement, an early post-assignment outcome, to serve as a separator.

The panels are intentionally not pooled. Each encodes a different statistical question, and including or excluding kindergarten achievement changes the meaning of separation between assignment and the later outcome.

5.4 Variable coding

Table 5.2: Variables supplied to FCI and FCI+.

Analysis variable	Source fields	Coding
K_Class	gkclasstype	Small, regular, or regular with aide.
Gender	gender	Male or female, using the official data codes.
Ethnicity	race	White, Black, or Other; smaller source categories are collapsed to avoid extremely sparse contingency cells.
Free_Lunch	gkfreelunch	Free/reduced-price lunch versus non-free lunch.
School_Context	gksurban	Inner city, suburban, rural, or urban.
Entry_Age	birthyear, birthmonth	Age in months at September 1985, converted to four quantile categories.
Teacher_Experience	gktyears	0–5, 6–10, 11–20, or 21+ years.
K_Achievement	gktreadss, gktmathss	Reading plus mathematics scaled score, converted to kindergarten achievement quartiles.
Grade3_Achievement	g3treadss, g3tmathss	Reading plus mathematics scaled score, converted to grade-3 achievement quartiles among students with both scores.
Grade3_Observed	g3treadss, g3tmathss	Indicator that both grade-3 scores are present.

All model inputs are integer-coded discrete categories. The exact numeric tables supplied to the algorithms are committed under `case_studies/tennessee_star/data/processed`.

STAR Analysis Methods

6.1 Descriptive randomized-arm reference

Before causal discovery, the report computes observed means and randomized-arm contrasts for:

1. the sum of kindergarten reading and mathematics scaled scores;
2. the sum of grade-3 reading and mathematics scaled scores among students with both scores;
3. the probability that both grade-3 scores are observed.

Uncertainty is estimated with 1,000 bootstrap samples of kindergarten schools. Resampling schools rather than individual students preserves the main within-school dependence in the descriptive comparison. These contrasts are reported separately from FCI output because they use the experiment's assignment as the basis for interpretation.

6.2 Discrete G-square conditional-independence test

The application uses the likelihood-ratio G-square test. For variables X and Y , conditioning set S , observed contingency counts O , and expected counts E under $X \perp Y \mid S$, the statistic is

$$G^2 = 2 \sum_s \sum_{x,y} O_{xys} \log \left(\frac{O_{xys}}{E_{xys}} \right),$$

with zero-count contributions omitted. The test is appropriate for the discretised panels, but high-order conditioning can create sparse cells. That is one reason the FCI⁺ bound is fixed at $k = 3$ and why sensitivity is reported rather than hidden.

6.3 Primary discovery configuration

Table 6.1: Primary STAR discovery configuration.

Setting	Value
CI test	G-square on discretised variables
Significance threshold	$\alpha = 0.05$
Standard FCI	<code>profile="paper"</code> ; unbounded paper schedule
Paper FCI ⁺	<code>profile="paper"</code> ; $k = 3$
Robust application FCI ⁺	<code>profile="practical"</code> ; stable skeleton; strongest separating set; conservative colliders; robust orientation; depth and sparsity bound 3
R reference	<code>pcalg::fciPlus</code> ; selection bias enabled; no exposed explicit k setting
Timing	Three full-data repetitions; median elapsed time reported
PAG stability	12 kindergarten-school cluster bootstrap fits per algorithm and panel for all three self-implemented profiles
Order audit	Every cyclic column shift for all local profiles and the R reference
Random seed	Fixed by the application script for reproducibility
Prior temporal constraints	None supplied to discovery; temporal reversals are audited after fitting

Not imposing temporal or randomization constraints allows the analysis to test the behaviour of the data-only algorithm. It also means that implausible orientations must not be treated as credible causal chronology.

6.4 School-cluster PAG stability

For each panel and algorithm, the 79 kindergarten schools are sampled with replacement. All student rows from a selected school are included, including repeated copies when a school is sampled more than once. FCI or FCI⁺ is then refitted and the presence of each unordered adjacency is recorded.

With $B = 12$ replicates, the estimated adjacency frequency is

$$\hat{\pi}_{XY} = \frac{1}{B} \sum_{b=1}^B \mathbf{1}\{X \text{ and } Y \text{ are adjacent in replicate } b\}.$$

This is an exploratory stability diagnostic. Twelve replicates give a coarse grid of 8.3 percentage points and are not sufficient for precise tail probabilities.

6.5 Sensitivity analysis

The focused treatment-outcome analysis is repeated for:

- $\alpha \in \{0.01, 0.05\}$;
- three versus four quantile bins for age and achievement;

- standard FCI, paper-profile FCI⁺, and robust application FCI⁺.

The target is whether `K_Class` remains adjacent to `Grade3_Achievement`. This directly tests whether the most substantively tempting PAG edge is robust to reasonable analysis choices.

6.6 Decision rule for robust conclusions

The application distinguishes a reproducible adjacency from a credible direction. A skeleton-level conclusion is promoted only when the target adjacency survives every cyclic column shift and has high school-cluster bootstrap frequency. A directional conclusion additionally requires invariant endpoint marks, no conflict with measurement time or randomization, and no material reversal under the alpha/binning sensitivity checks.

Failure of the directional rule does not delete the edge. It downgrades the claim to conditional-dependence structure with unresolved direction. This pre-specified rule is why the robust profile's circle endpoints are treated as a reliability improvement rather than an incomplete result.

STAR Results

7.1 Observed randomized-arm outcomes

Table 7.1: Observed outcome summaries by kindergarten class arm. Intervals use 1,000 kindergarten-school bootstrap samples.

Outcome	Arm	<i>n</i>	Estimate	95% interval
Kindergarten score	Regular	2,005	918.04	908.78 to 927.05
	Regular + aide	2,043	918.36	908.52 to 927.07
	Small	1,738	931.94	922.70 to 940.12
Grade-3 score	Regular	1,047	1246.74	1238.96 to 1254.01
	Regular + aide	1,019	1247.25	1238.53 to 1255.48
	Small	936	1258.43	1250.58 to 1266.17
Grade-3 observed	Regular	2,194	47.72%	42.72% to 52.14%
	Regular + aide	2,231	45.67%	40.64% to 50.39%
	Small	1,900	49.26%	44.46% to 53.77%

Grade-3 score summaries condition on both reading and mathematics scores being observed. Kindergarten score summaries also exclude missing score components, so they are design-supported observed-outcome comparisons rather than a fully reconstructed intention-to-treat estimand.

Randomized-arm descriptive contrasts with 95% school-cluster intervals

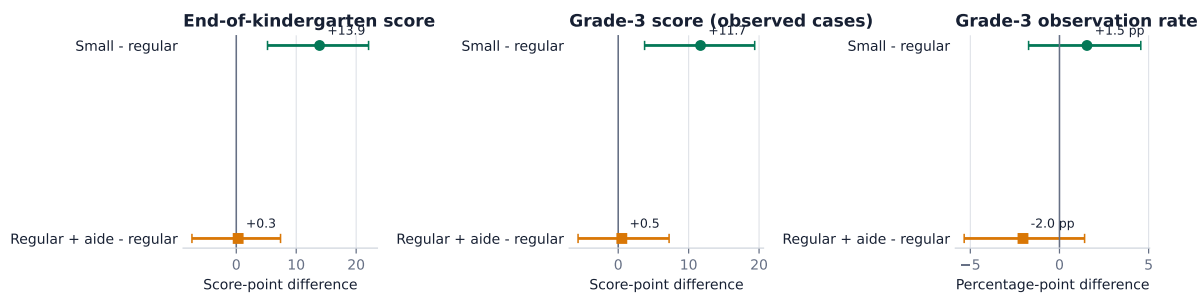


Figure 7.1: Contrasts against the regular-class arm. Small classes show positive score differences at kindergarten and grade 3 among observed cases; the aide arm remains close to zero.

The small-minus-regular kindergarten contrast is 13.90 points, with a 95% school-cluster interval from 5.21 to 22.07. The regular-with-aide contrast is 0.31 points, with an interval from -7.40 to 7.38 .

At grade 3 among observed students, the small-class contrast remains positive at 11.69 points (3.74 to 19.38). The aide contrast is 0.51 points (-5.70 to 7.22). Observation rates themselves do not show a precise class-arm difference: small minus regular is $+1.54$ percentage points (-1.73 to 4.56), while aide minus regular is -2.05 points (-5.35 to 1.41).

Key result. The observed randomized-arm evidence supports a small-class achievement advantage and does not support a comparable advantage from adding an aide to a regular class. The grade-3 score contrast is conditional on later observation and therefore has a weaker causal interpretation than the early outcome comparison.

7.2 Attrition and grade-3 observation

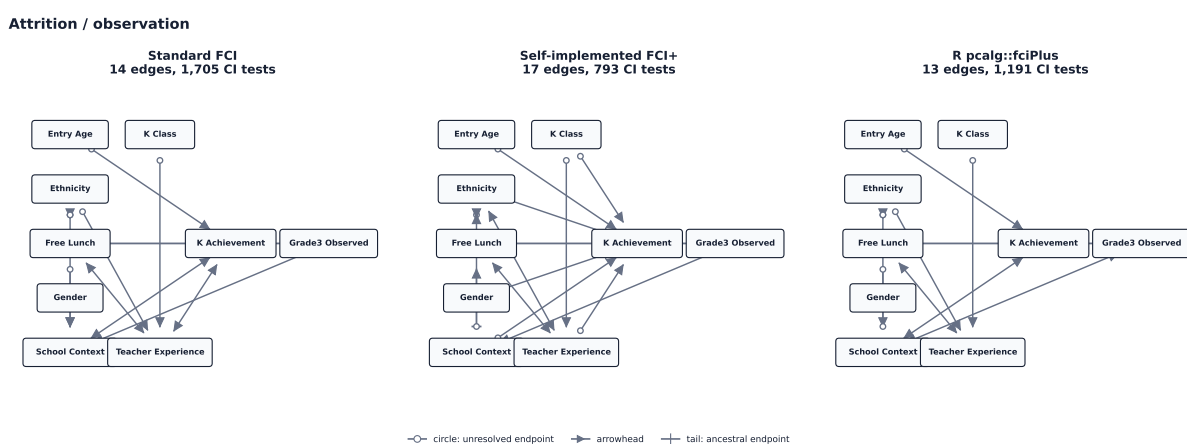


Figure 7.2: Three PAGs for the attrition/observation panel: self-implemented FCI, self-implemented FCI⁺, and R `pcalg::fciPlus`. The skeletons substantially overlap, while the achievement/observation endpoint differs.

Standard FCI returns 14 edges after 1,705 CI tests; self-implemented FCI⁺ returns 17 edges after 793 tests; and R `pcalg::fciPlus` returns 13 edges after 1,191 tests. The FCI/self-FCI⁺, FCI/R-FCI⁺, and

self-/R-FCI⁺ skeleton Jaccard values are 0.824, 0.929, and 0.765. All three omit a direct `K_Class`–`Grade3_Observed` adjacency. In the self-implementation school bootstrap that adjacency appears in only 17% of FCI fits and 8% of both paper and robust FCI⁺ fits.

By contrast, `K_Achievement`–`Grade3_Observed` is adjacent in 100% of the bootstrap fits for all three local profiles. The corresponding free-lunch/observation adjacency is also stable in the targeted audit. These results suggest that later data availability is not independent of early achievement and measured socioeconomic or school context.

The three endpoint readings are revealing. Standard FCI gives the impossible *grade-3 observation* → *kindergarten achievement*; self-FCI⁺ leaves a circle at the later endpoint; and R-FCI⁺ gives the temporally admissible *kindergarten achievement* → *grade-3 observation*. The R result is more plausible on this edge, but package identity alone cannot establish that its orientation is statistically correct.

Interpretation caution. The attrition PAG does not prove a particular missingness mechanism. It shows that the observed conditional-independence structure links grade-3 availability to earlier achievement and context. A formal missing-data analysis would require an explicit estimand and additional assumptions.

7.3 Longitudinal achievement

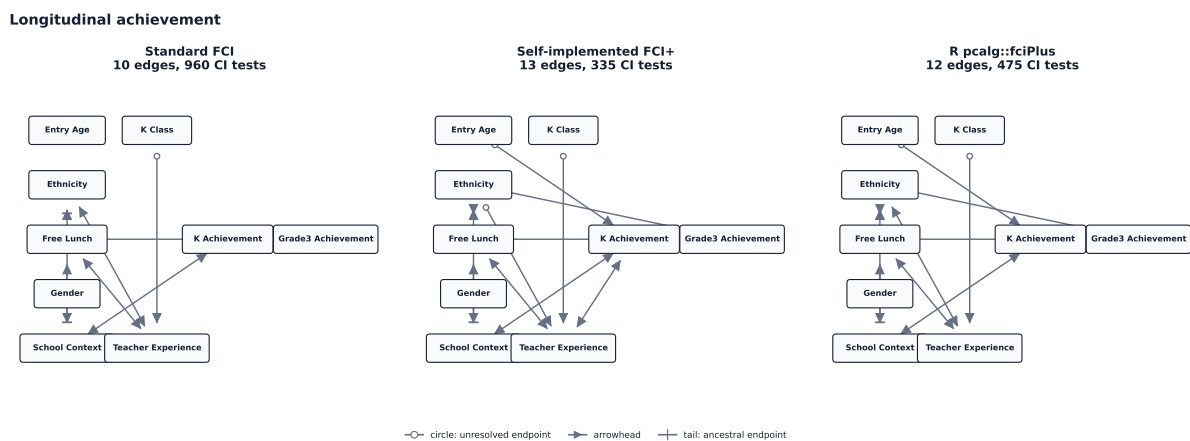


Figure 7.3: Three PAGs for the longitudinal complete-case panel. All retain the kindergarten/grade-3 achievement adjacency, but the two FCI⁺ implementations orient it against chronology.

Standard FCI returns 10 edges and 960 CI tests. Self-FCI⁺ returns 13 edges and 335 tests, while R-FCI⁺ returns 12 edges and 475 tests. Pairwise skeleton Jaccard is 0.769 for FCI/self-FCI⁺, 0.833 for FCI/R, and 0.923 for the two FCI⁺ implementations.

All three algorithms retain the kindergarten/grade-3 achievement adjacency, and it appears in all 12 school-cluster bootstrap samples for all three local profiles. Neither full-data PAG retains a direct class-assignment/grade-3 achievement adjacency once kindergarten achievement is included. That treatment-outcome adjacency nevertheless reappears in 50% of FCI and 67% of FCI⁺ paper-profile bootstrap fits. It appears in all robust-profile bootstrap fits despite being absent from the robust full-data PAG. This discordance is a warning that conservative orientation and order invariance do not remove school-resampling sensitivity.

Both FCI⁺ implementations orient grade-3 achievement toward kindergarten achievement, which is temporally impossible if read causally. Standard FCI leaves the grade-3 endpoint unresolved. The agreement between two implementations is therefore not sufficient to override chronology: it may replicate the same finite-sample CI pattern and orientation logic under complete-case selection.

7.4 Focused treatment-outcome panel

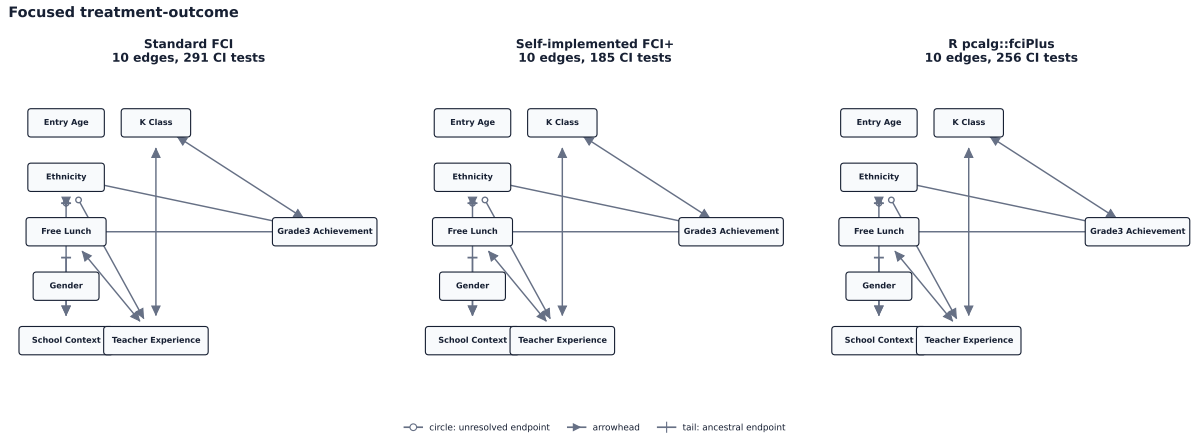


Figure 7.4: PAGs for the focused complete-case panel. All three algorithms recover the exact same 10-edge skeleton and endpoint pattern, including the bidirected class-assignment/grade-3 edge.

Removing kindergarten achievement from the node set changes the result. Standard FCI, self-FCI⁺, and R-FCI⁺ all return the same 10-edge PAG, not merely the same skeleton. All three contain

$$K_Class \leftrightarrow Grade3_Achievement.$$

The adjacency appears in 92% of the FCI school bootstraps, 83% of the paper self-FCI⁺ bootstraps, and 100% of the robust-profile bootstraps. The independent exact PAG replication makes this observed conditional-independence pattern more credible, but does not make the literal latent-confounding interpretation credible.

A naive reading would call the bidirected edge evidence of latent confounding. That reading is not defensible here. Kindergarten class assignment was randomized, while the focused panel conditions on grade-3 outcomes being observed and omits early post-assignment achievement. Conditioning on observation, school-level dependence, omitted post-randomization variables, and CI-test approximation can all create an observed independence pattern that is represented by arrowheads at both endpoints.

7.5 Robust application profile

Robust FCI+ application profile: conservative STAR PAGs

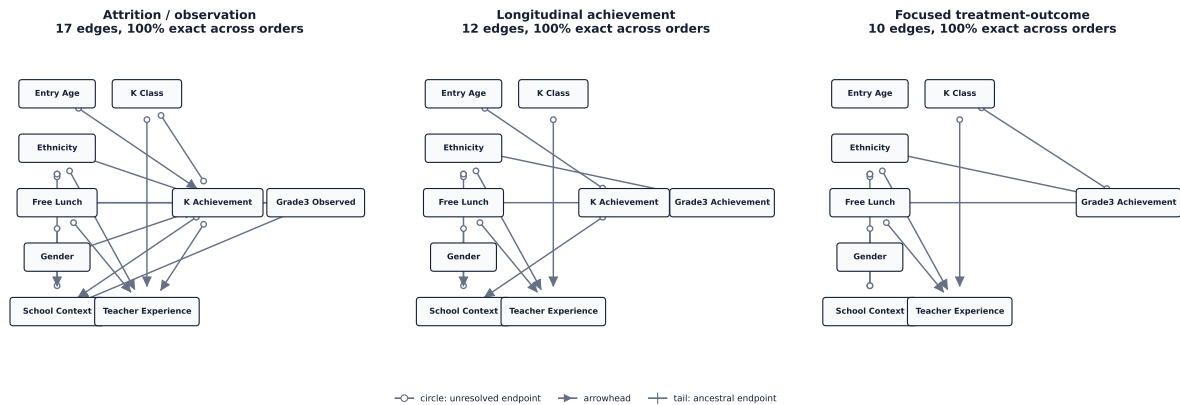


Figure 7.5: The three STAR PAGs from the separate robust application profile. Stable skeleton search and conservative orientation preserve the main adjacencies while leaving unsupported directions unresolved.

The robust profile is not presented as a fourth paper implementation. It is a finite-sample decision layer applied after the literal FCI and FCI⁺ comparisons. It returns 17, 12, and 10 edges on the attrition, longitudinal, and focused panels, respectively. The target relations are

$$\begin{aligned} K_Achievement &\circ\circ Grade3_Observed, \\ K_Achievement &\circ\circ Grade3_Achievement, \\ K_Class &\circ\circ Grade3_Achievement. \end{aligned}$$

Thus it retains the relevant skeleton information without repeating the paper-profile chronology reversal or the focused-panel bidirected over-interpretation.

Table 7.2: Cyclic-order audit for paper and robust self-FCI⁺ profiles.

Panel	Profile	Exact PAG	Worst skeleton	Temporal reversals
Attrition	Paper	100.0%	100.0%	0
	Robust	100.0%	100.0%	0
Longitudinal	Paper	55.6%	92.3%	1
	Robust	100.0%	100.0%	0
Focused	Paper	75.0%	100.0%	0
	Robust	100.0%	100.0%	0

Key result. The robust profile improves the measured reliability of the STAR output: every tested column order returns the same complete PAG, every target adjacency survives, and no fully directed edge runs backward in measurement time. The price is deliberate under-orientation and additional CI work. Reliability has improved; identifiability has not.

7.6 Stability and sensitivity

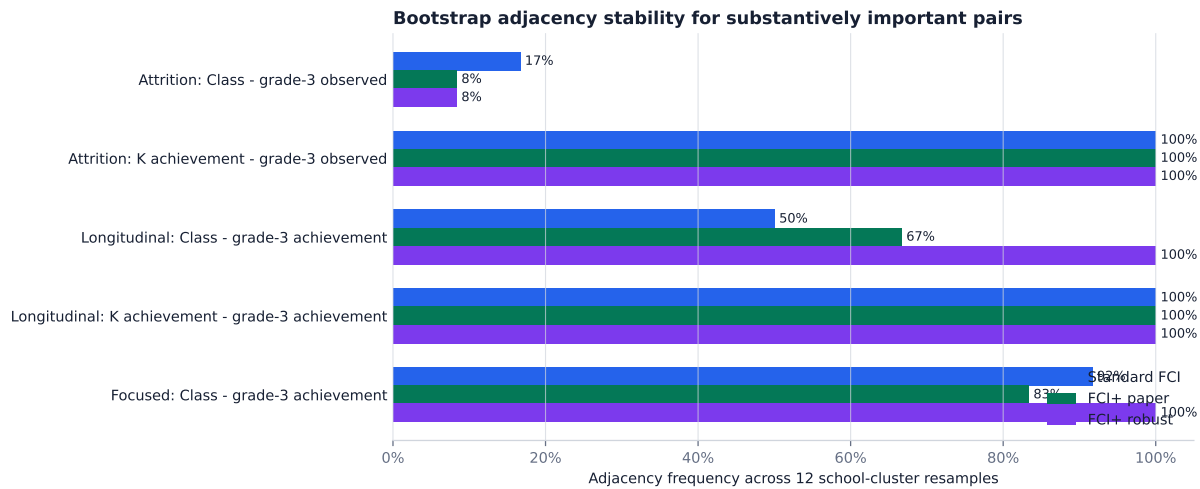


Figure 7.6: School-cluster adjacency frequencies for selected pairs. Frequencies measure skeleton stability only; they do not validate endpoint orientation.

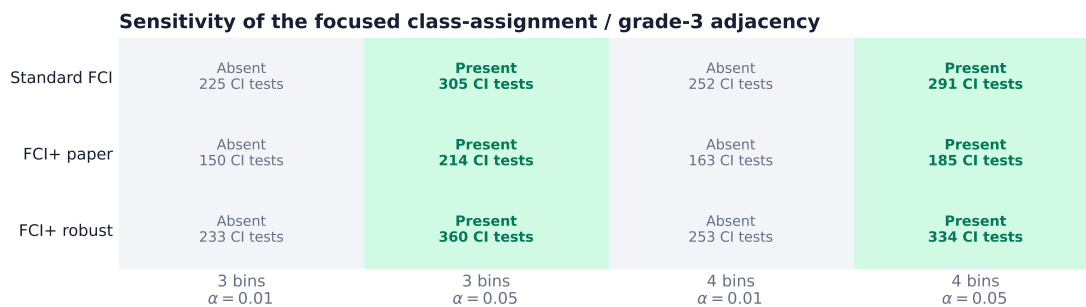


Figure 7.7: Focused treatment-outcome sensitivity. The adjacency appears at $\alpha = 0.05$ and disappears at $\alpha = 0.01$, regardless of whether three or four quantile bins are used.

The sensitivity result is decisive for interpretation. The focused treatment-outcome adjacency is robust to the tested bin count but not to the CI threshold. At $\alpha = 0.01$, none of the three local profiles retains it; at $\alpha = 0.05$, the paper profiles retain a bidirected mark while the robust profile retains the adjacency with two unresolved circles.

The stronger threshold requires more evidence against conditional independence before an edge is kept. Its removal therefore indicates that the edge is driven by relatively weak conditional-dependence evidence, not an invariant feature across reasonable test settings.

7.7 Cross-implementation agreement and order audit

Cross-implementation agreement is stronger for skeletons than endpoints

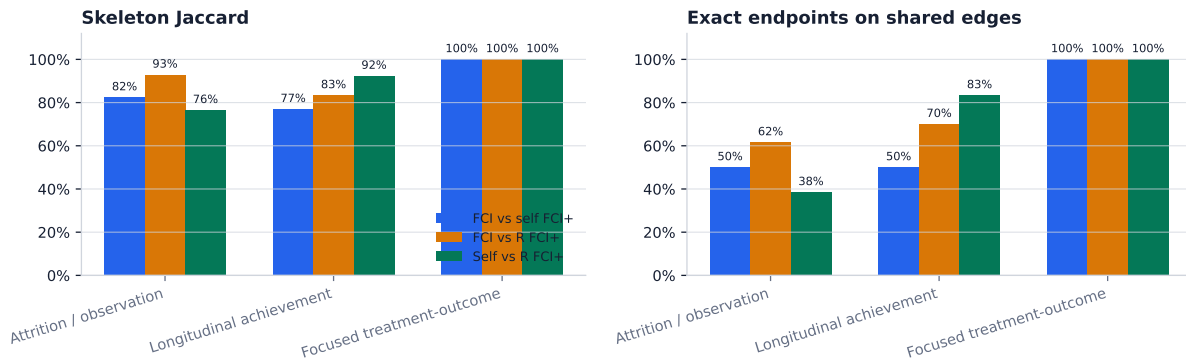


Figure 7.8: Pairwise skeleton Jaccard and exact endpoint agreement on shared edges. Skeleton agreement is consistently stronger than endpoint agreement outside the focused panel.

Table 7.3: Three-algorithm structural agreement. Endpoint rates use only adjacencies shared by the indicated pair.

Panel	Pair	Skeleton Jaccard	Exact endpoint rate
Attrition	FCI / self-FCI ⁺	0.824	50.0%
	FCI / R-FCI ⁺	0.929	61.5%
	self- / R-FCI ⁺	0.765	38.5%
Longitudinal	FCI / self-FCI ⁺	0.769	50.0%
	FCI / R-FCI ⁺	0.833	70.0%
	self- / R-FCI ⁺	0.923	83.3%
Focused	FCI / self-FCI ⁺	1.000	100.0%
	FCI / R-FCI ⁺	1.000	100.0%
	self- / R-FCI ⁺	1.000	100.0%

Cyclic variable-order robustness separates skeletons from endpoints

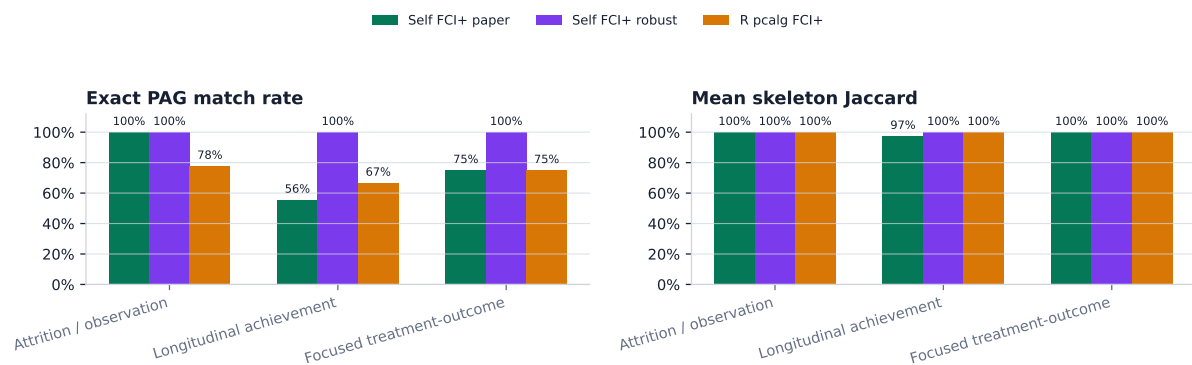


Figure 7.9: Cyclic-order audit for paper self-FCI⁺, robust self-FCI⁺, and R pcalg::fciPlus. Skeleton agreement is generally stronger than exact endpoint agreement; the robust profile is invariant on both measures.

The order audit rotates the input variable list through every cyclic shift. The R reference preserves every panel skeleton, while reproducing exact PAG endpoints in 7 of 9 attrition orders, 6 of 9 longitudinal orders, and 6 of 8 focused orders. Paper self-FCI⁺ is exact in 9 of 9, 5 of 9, and 6 of 8 orders, respectively; its longitudinal skeleton changes slightly in some orders. Robust self-FCI⁺ is exact in all 26 refits. The combined evidence supports skeleton robustness while independently showing that more aggressive endpoint marks can be implementation- and order-sensitive.

7.8 Computational comparison

STAR fits: paper validation and robust application profile

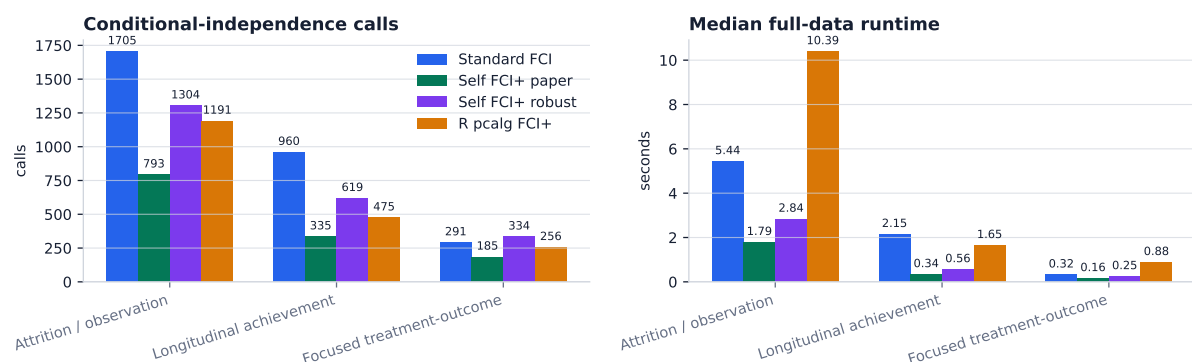


Figure 7.10: Full-data search work and median runtime. All algorithms use the same panel, compact-table G-square test, and $\alpha = 0.05$; internal search schedules and language runtimes are not identical.

Table 7.4: Paper-comparison and robust-profile performance on the STAR panels.

Panel	Algorithm	Edges	CI tests	Median seconds
Attrition	FCI	14	1,705	5.437
	Self-FCI ⁺ paper	17	793	1.788
	Self-FCI ⁺ robust	17	1,304	2.838
	R-FCI ⁺	13	1,191	10.394
Longitudinal	FCI	10	960	2.148
	Self-FCI ⁺ paper	13	335	0.337
	Self-FCI ⁺ robust	12	619	0.562
	R-FCI ⁺	12	475	1.650
Focused	FCI	10	291	0.322
	Self-FCI ⁺ paper	10	185	0.159
	Self-FCI ⁺ robust	10	334	0.247
	R-FCI ⁺	10	256	0.884

FCI⁺ uses 46.5%, 34.9%, and 63.6% of the standard FCI CI-test counts on the attrition, longitudinal, and focused panels. The independent R FCI⁺ uses 69.9%, 49.5%, and 88.0% of the corresponding standard-FCI counts. Runtime is not a language-neutral benchmark: the R runner includes R-level contingency-table construction, whereas the Python implementation uses the package's native data path. CI-call counts are the more interpretable search-work comparison.

Relative to paper self-FCI⁺, the robust profile uses 1.64, 1.85, and 1.81 times as many CI calls. Relative to standard FCI, it uses 76.5%, 64.5%, and 114.8%. The added work purchases stable skeleton deletion and invariant conservative endpoints; it is not claimed as a speed improvement.

These results demonstrate reduced search work, not universal algorithmic dominance. In the two panels where skeleton Jaccard is below one, efficiency and empirical structure differ simultaneously.

Causal Interpretation

8.1 What can be concluded

8.1.1 Small classes improve early observed achievement

STAR's random assignment is the primary causal identification mechanism. Within the kindergarten cohort and among students with observed score components, the small-class arm is 13.9 points above the regular arm, with a school-cluster interval that excludes zero. This supports a beneficial causal effect of smaller classes on early achievement, subject to outcome missingness, assignment compliance, and the use of the observed composite score.

The PAG is consistent with an assignment/achievement connection in the FCI⁺ attrition panel, but the causal conclusion does not depend on that endpoint. It depends on the experiment.

8.1.2 Adding an aide does not show the same advantage

The regular-with-aide contrasts are close to zero at kindergarten and grade 3, and both intervals include zero. In these observed data there is no evidence that adding an aide to a regular-size class reproduces the achievement advantage of reducing class size.

8.1.3 Early achievement strongly predicts later structure

Kindergarten and grade-3 achievement remain adjacent in the full longitudinal fits of all three algorithms and in 100% of the self-implementation school bootstraps. Kindergarten achievement is also adjacent to grade-3 observation in all three attrition fits. Cross-implementation agreement therefore supports a robust *skeleton-level* relationship between early performance, later performance, and later data availability.

This finding still does not quantify mediation. Kindergarten achievement is a post-assignment variable, and a formal mediation analysis would require additional counterfactual assumptions. Moreover, the direction of the longitudinal achievement edge fails the chronological audit for both FCI⁺ implementations.

8.1.4 Attrition is selective but not clearly caused by class arm

The full-data attrition PAGs do not retain a direct class-arm/observation adjacency, and its bootstrap frequency is low. Observation is more stably connected to early achievement and measured context. The

evidence therefore supports concern about selective attrition without showing a stable direct effect of class arm on grade-3 observation.

8.2 What cannot be concluded

1. **No unique DAG is identified.** Every PAG represents an equivalence class.
2. **No causal effect magnitude comes from FCI.** The score differences are arm contrasts; PAG edges contain no “points gained” coefficient.
3. **A bidirected edge is not proof of a hidden confounder.** In the focused panel it conflicts with the known randomization unless selection, post-treatment conditioning, or other violations are considered.
4. **The grade-3 contrast is not a complete long-run intention-to-treat estimate.** It conditions on observed grade-3 outcomes.
5. **Temporal reversals are not credible chronology.** They are algorithm diagnostics under violated or approximate assumptions.
6. **Implementation agreement is not proof.** Shared CI decisions and shared logical rules can reproduce the same wrong endpoint.

Key result. The strongest causal conclusion is design-based: smaller kindergarten classes improve early observed achievement relative to regular classes, while the aide arm shows no comparable effect. The main contribution of the three-algorithm comparison is to separate replicated skeleton structure from unstable endpoint orientation. It exposes selection, persistence, and specification sensitivity around the experiment; it does not replace it.

Critical Self-Assessment

9.1 Strengths

- The algorithm and application are physically and logically separated.
- The data source, user guide, exact model inputs, random seeds, result tables, and report artifacts are committed.
- Standard FCI is always reported alongside FCI⁺.
- An independently maintained R implementation is run with matched CI semantics and its disagreements are retained in the report.
- Randomized-arm evidence is separated from exploratory PAG evidence.
- School-level resampling acknowledges the most important clustering unit.
- Endpoint reversals and threshold sensitivity are reported as failures of literal interpretation rather than silently corrected.
- The software is validated with exact oracle graphs, finite-sample integration tests, strict typing, multi-version tests, and wheel installation checks.

9.2 Statistical limitations

9.2.1 Discretisation and sparse contingency tables

Achievement and age are reduced to quantile categories, and teacher experience is binned. This enables G-square testing across mixed educational variables but discards within-bin information. As conditioning sets grow, some multiway contingency cells may have small expected counts, weakening the chi-square approximation.

9.2.2 Dependence within schools and classrooms

The primary CI test treats student rows as independent. The bootstrap resamples schools, which measures sensitivity to school composition, but it does not turn each underlying G-square test into a cluster-robust CI test. Classroom identifiers and multilevel structure should be incorporated in a more specialised analysis.

9.2.3 Complete-case selection

The longitudinal and focused panels require grade-3 scores. If observation is affected by prior achievement, mobility, family circumstances, or school processes, conditioning on complete cases may open selection paths. The attrition panel provides evidence that observation is structurally related to early variables, reinforcing this concern.

9.2.4 Post-treatment variables

Kindergarten achievement and teacher experience are measured during the treatment period. They are not baseline covariates in the same sense as sex or ethnicity. Including kindergarten achievement may block part of the class-size pathway; excluding it may leave omitted post-treatment dependence. The two panels deliberately show both specifications instead of selecting the one with the preferred edge.

9.2.5 Faithfulness and finite-sample CI error

FCI-family guarantees rely on faithfulness and correct CI decisions. Near-zero or cancelling effects may violate faithfulness, while finite samples produce false dependence and false independence. The threshold-sensitive focused edge is a direct example of this limitation.

9.2.6 Unknown sparsity bound

The FCI⁺ paper profile uses $k = 3$. The true maximum degree of the underlying STAR MAG is not observed and cannot be verified from the learned graph alone. If the required separating set exceeds the bound, FCI⁺ may retain an extra edge. Conversely, `pcalg::fciPlus` does not expose a directly comparable k argument, so the independent reference is not an identical internal search schedule.

9.2.7 Endpoint order sensitivity

The R and paper-profile self-FCI⁺ audits change some endpoint marks; the paper self-FCI⁺ longitudinal skeleton also changes slightly. This is a practical warning for constraint-based orientation under finite-sample CI decisions: deterministic output for one column order is not the same as order-invariant causal information. The robust profile removes the observed order dependence by stable deletion and conservative orientation. The report therefore uses it for applied interpretation, while still giving more evidential weight to skeleton consensus than to isolated endpoint marks.

9.2.8 Coarse PAG bootstrap

Twelve graph bootstrap replicates are computationally useful but statistically coarse. Frequencies such as 83% and 92% correspond to ten and eleven successful replicates. They should be viewed as stability indicators, not precise probabilities that an edge is true.

9.3 Substantive limitations

Limitation. The STAR application demonstrates an auditable causal-structure workflow; it is not a definitive re-estimation of the experimental treatment effect, and known-impossible orientations are reported as diagnostics.

Original randomization blocks, class switching, compliance, and school-specific assignment procedures are not reconstructed; a policy estimate must return to those records and define an intention-to-treat estimand. The transparent composite score simply adds reading and mathematics, leaving test scaling, subject-specific, distributional, and subgroup effects unexamined.

Conclusions and Future Work

10.1 Conclusions

The project achieved its two-stage objective.

First, it produced an integrated and auditable implementation of standard FCI and FCI⁺. Exact m-separation tests reproduce published PAGs, including the hierarchical removal of the Figure 4(b) D-SEP link. The package exposes the algorithm through a concise API while retaining separating sets, CI traces, orientation traces, and assumption notes. FCI⁺ materially reduces CI-test work in the oracle fixture and in every STAR panel.

Second, it applied the package to a real educational experiment without mixing domain code into the algorithm. The STAR results show a clear observed small-class achievement advantage, no comparable aide advantage, stable early-to-later achievement structure, and evidence that later observation is selective. An independent R `pcalg::fciPlus` run exactly replicates the focused 10-edge PAG and broadly supports the other skeletons. It also shows why PAGs require disciplined interpretation: the most tempting bidirected treatment-outcome edge is conditional on complete cases and disappears under a stricter alpha, while some endpoint marks change with variable order or violate chronology.

A separate robust application profile then addresses the principal empirical failure found by those audits. Across every cyclic variable order it returns the same PAG in all three panels, preserves the main target adjacencies, and reports no fully directed temporal reversal. It does so by leaving unsupported endpoints unresolved and by spending more CI calls. This is the appropriate robustness trade-off for the application: fewer directional claims, backed by stronger reproducibility.

No algorithm is uniformly “best” on the STAR data. The paper-profile self-FCI⁺ is the most auditable validation implementation in this repository, passes the exact oracle fixtures, and usually requires the fewest CI calls. The robust self-FCI⁺ profile is the preferred empirical output because it passes the order and temporal audits, although its circles identify less direction. R `pcalg` provides valuable independent replication and gives the most plausible direction for the attrition target edge, yet shares the longitudinal reversal and exhibits endpoint order sensitivity. Standard FCI is less economical but leaves the longitudinal target partly unresolved rather than fully reversing it. Credibility must therefore be assigned claim by claim.

The report therefore rejects two extremes. It is too pessimistic to conclude that causal discovery adds nothing to a randomized study; the attrition and stability analysis reveals useful structure. It is too optimistic to conclude that FCI has rediscovered and fully identified the experiment’s causal model.

The correct contribution lies between them: an auditable map of conditional independence constraints, uncertainty, and modelling sensitivity.

10.2 Recommended extensions

1. Reconstruct the randomization strata, estimate formal intention-to-treat effects, and model grade-3 observation explicitly.
2. Develop cluster-aware or multilevel CI tests for students nested within classrooms and schools.
3. Increase PAG bootstrap repetitions and report Monte Carlo uncertainty for stability frequencies.
4. Run separate reading and mathematics analyses and examine prespecified treatment-effect heterogeneity.
5. Evaluate defensible temporal background knowledge separately from the data-only robust profile, without silently forcing preferred arrows.
6. Compare alternative CI tests and $k = 2, 3, 4$ FCI⁺ profiles while diagnosing which edges depend on the sparsity bound.
7. Increase bootstrap repetitions and extend stability reporting to endpoint compatibility, rather than only adjacency frequencies.

Reproducibility

A.1 Repository layout

Path	Contents
<code>src/fci_engine/</code>	Reusable FCI and FCI ⁺ package.
<code>tests/</code>	Unit, oracle, integration, typing, and report tests.
<code>case_studies/tennessee_star/download_data.py</code>	Official Harvard Dataverse downloader and shape validation.
<code>case_studies/tennessee_star/study.py</code>	STAR cohort construction, coding, discovery, bootstrap, sensitivity, and summary payload.
<code>case_studies/tennessee_star/report.py</code>	Standalone HTML report renderer.
<code>case_studies/tennessee_star/pcalg_reference.R</code>	Independent CRAN <code>pcalg::fciPlus</code> runner and matched G-square CI test.
<code>case_studies/tennessee_star/pcalg_reference.py</code>	R runner integration, artifact loading, and metadata validation.
<code>case_studies/tennessee_star/output/</code>	Machine-readable PAG, benchmark, bootstrap, sensitivity, contrast, and visual report artifacts.
<code>reports/fci_plus_star_report.tex</code>	This complete LaTeX report.
<code>reports/generate_report_figures.py</code>	Vector-figure generator reading the committed result JSON.

A.2 Commands

Listing A.1: Reproduce the STAR application.

```
PYTHONPATH=src python -m \
  case_studies.tennessee_star.run_case_study \
  --benchmark-repeats 3 \
  --bootstraps 12 \
  --descriptive-bootstraps 1000 \
  --n-jobs 4
```

Listing A.2: Refresh the independent R reference and rebuild the integrated application.

```
PYTHONPATH=src python -m \
  case_studies.tennessee_star.run_case_study \
  --refresh-pcalg \
  --rscript /path/to/Rscript \
  --benchmark-repeats 3 \
  --bootstraps 12 \
  --descriptive-bootstraps 1000 \
  --n-jobs 4
```

Listing A.3: Regenerate the report figures.

```
PYTHONPATH=src python reports/generate_report_figures.py
```

Listing A.4: Run the software quality gates.

```
python -m pytest -q
python -m ruff check src tests examples case_studies reports
python -m mypy
python -m mypy case_studies/tennessee_star
python -m build --wheel
```

A.3 Data integrity

The committed official data files are:

- `STAR_Students.tab.gz`, a deterministic gzip of the Dataverse tabular export, SHA-256:

```
2ffa578822eb30fcd6626fa7c5cc734a
721fd05dd522403f0d643f89e891bac8
```

- `starUsersGuide.pdf`, SHA-256:

```
e51ff1d28d5af28c128196b3a133957
f9f2cd872b1abe348f156022193550130
```

Key STAR Result Tables

B.1 Primary discovery results

Table B.1: Primary full-data discovery results and temporal audit.

Panel	Algorithm	Samples	Edges	CI tests	Seconds	Temporal flag
Attrition	FCI	5,744	14	1,705	5.437	Grade3 Observed $\rightarrow$ K Achievement
Attrition	Self-FCI ⁺ paper	5,744	17	793	1.788	None
Attrition	Self-FCI ⁺ robust	5,744	17	1,304	2.838	None
Attrition	R-FCI ⁺	5,744	13	1,191	10.394	None
Longitudinal	FCI	2,787	10	960	2.148	None
Longitudinal	Self-FCI ⁺ paper	2,787	13	335	0.337	Grade3 Achievement $\rightarrow$ K Achievement
Longitudinal	Self-FCI ⁺ robust	2,787	12	619	0.562	None
Longitudinal	R-FCI ⁺	2,787	12	475	1.650	Grade3 Achievement $\rightarrow$ K Achievement
Focused	FCI	2,976	10	291	0.322	None
Focused	Self-FCI ⁺ paper	2,976	10	185	0.159	None
Focused	Self-FCI ⁺ robust	2,976	10	334	0.247	None
Focused	R-FCI ⁺	2,976	10	256	0.884	None

B.2 Focused sensitivity results

Table B.2: Presence of the class-assignment/grade-3 adjacency.

Bins	α	Algorithm	Adjacent?	CI tests
3	0.01	FCI	No	225
3	0.01	FCI+ paper	No	150
3	0.01	FCI+ robust	No	233
3	0.05	FCI	Yes, $K_Class \leftrightarrow G3$	305
3	0.05	FCI+ paper	Yes, $K_Class \leftrightarrow G3$	214
3	0.05	FCI+ robust	Yes, $K_Class \leftarrow \circ G3$	360
4	0.01	FCI	No	252
4	0.01	FCI+ paper	No	163
4	0.01	FCI+ robust	No	253
4	0.05	FCI	Yes, $K_Class \leftrightarrow G3$	291
4	0.05	FCI+ paper	Yes, $K_Class \leftrightarrow G3$	185
4	0.05	FCI+ robust	Yes, $K_Class \circ - \circ G3$	334

Researcher Interpretation Checklist

Before treating an FCI-family result as substantive causal evidence, the following questions should be answered.

1. What is the estimand: a graph, an ancestry claim, or a causal effect?
2. Which variables are pre-treatment, post-treatment, outcomes, or selection indicators?
3. Does the chosen CI test match the data type and missingness pattern?
4. Are observations independent, or is clustering present?
5. Is the FCI⁺ sparsity bound plausible and sensitivity-tested?
6. Does standard FCI recover the same important skeleton relations?
7. Are adjacencies stable under resampling?
8. Are endpoints stable under alpha, binning, and variable-set changes?
9. Do any directed edges violate known temporal order or randomization?
10. Is a bidirected edge being over-interpreted as proof of a specific latent variable?
11. Are design-based conclusions being kept separate from data-only PAG conclusions?
12. Can every published number be regenerated from committed inputs?

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